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$$A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} -7 \\ 5 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & 2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

$LY = b$;

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -7 \\ -1 & 1 & 0 & 5 \\ 2 & -5 & 1 & 2 \end{array}\right] \begin{array}{l} \\ +R_1 \\ -2R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -2 \\ 0 & -5 & 1 & 16 \end{array}\right] \begin{array}{l} \\ \\ +5R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 6 \end{array}\right] \Rightarrow \begin{cases} Y_1 = -7 \\ Y_2 = -2 \\ Y_3 = 6 \end{cases}$$

$UX = Y$

$$\left[\begin{array}{ccc|c} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & -1 & 6 \end{array}\right] \begin{array}{l} -2R_3 \\ -R_3 \\ \\ \end{array} \left[\begin{array}{ccc|c} 3 & -7 & 0 & -19 \\ 0 & -2 & 0 & -8 \\ 0 & 0 & -1 & 6 \end{array}\right] \begin{array}{l} \\ /(-2) \\ /(-1) \end{array}$$

$$\left[\begin{array}{ccc|c} 3 & -7 & 0 & -19 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{array}\right] +7R_2 \left[\begin{array}{ccc|c} 3 & 0 & 0 & 9 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{array}\right] \Rightarrow \begin{cases} 3x_1 = 9 \\ x_2 = 4 \\ x_3 = -6 \end{cases}$$

$$LY = b \Rightarrow Y = \begin{bmatrix} -1 \\ -2 \\ 6 \end{bmatrix}, \quad UX = Y = X = \begin{bmatrix} 3 \\ 4 \\ -6 \end{bmatrix}$$

$\boxed{N\,3}$

$$A = \begin{bmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$LY = b$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ -3 & 1 & 0 & 0 \\ 4 & -1 & 1 & 4 \end{array}\right] \begin{array}{l} \\ +3R_1 \\ -4R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & -1 & 1 & 0 \end{array}\right] +R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & 3 \end{array}\right] \Rightarrow \begin{cases} y_1 = 1 \\ y_2 = 3 \\ y_3 = 3 \end{cases}$$

$$\left[\begin{array}{ccc|c} 2 & -1 & 0 & -5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 3 \end{array}\right] + R_2 \left[\begin{array}{ccc|c} 2 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 3 \end{array}\right] \quad X = \begin{cases} 2x_1 = -2 \\ x_2 = 3 \\ x_3 = 3 \end{cases}$$

[redacted]

$$LY \Rightarrow Y = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \quad UX = Y \Rightarrow X = \begin{bmatrix} -1 \\ 3 \\ 3 \end{bmatrix}$$

$\boxed{N\ 11}$

$$\begin{bmatrix} 3 & -6 & 3 \\ 6 & -7 & 2 \\ -1 & 7 & 0 \end{bmatrix} \begin{array}{l} \\ -2R_1 \\ +R_1/3 \end{array} \begin{bmatrix} 3 & -6 & 3 \\ 0 & 5 & -4 \\ 0 & 5 & 1 \end{bmatrix} \begin{array}{l} \\ \\ -R_2 \end{array} \begin{bmatrix} 3 & -6 & 3 \\ 0 & 5 & -4 \\ 0 & 0 & 5 \end{bmatrix}$$

$$U = \begin{bmatrix} 3 & -6 & 3 \\ 0 & 5 & -4 \\ 0 & 0 & 5 \end{bmatrix} \quad L = \begin{bmatrix} \frac{3}{3} & 0 & 0 \\ \frac{6}{3} & \frac{5}{5} & 0 \\ -\frac{1}{3} & \frac{5}{5} & \frac{5}{5} \end{bmatrix} =$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -\frac{1}{3} & 1 & 1 \end{bmatrix}$$

$$LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -\frac{1}{3} & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -6 & 3 \\ 0 & 5 & -4 \\ 0 & 0 & 5 \end{bmatrix}$$

$\boxed{N\ 13}$

$$\begin{bmatrix} 1 & 3 & -5 & -3 \\ -1 & -5 & 8 & 4 \\ 4 & 2 & -5 & -7 \\ -2 & -4 & 7 & 5 \end{bmatrix} \begin{matrix} \\ +R_1 \\ -4R_1 \\ +2R_1 \end{matrix} \quad \begin{bmatrix} 1 & 3 & -5 & -3 \\ 0 & -2 & 3 & 1 \\ 0 & -10 & 15 & 5 \\ 0 & 2 & -3 & -1 \end{bmatrix} \begin{matrix} \\ \\ -5R_2 \\ +R_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & -5 & -3 \\ 0 & -2 & 3 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & -2 & 0 & 0 \\ 4 & -10 & 1 & 0 \\ -2 & 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 4 & 5 & 1 & 0 \\ -2 & -1 & 0 & 1 \end{bmatrix}$$

$$LU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 4 & 5 & 1 & 0 \\ -2 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & -5 & -3 \\ 0 & -2 & 3 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$